Applied Regression Analysis Using R

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Workshop Agenda

- Introduction to R
- Assessing the Distribution of Data
- Linear Regression
- Robust Regression
- Negative Binomial Regression
- GLM Gamma Regression
- *Break*
- Logistic Regression
- Linear Mixed Models
- Statistical Rigor
- One-on-one instruction, questions, etc.

Workshop Files

https://github.com/mikemalekahmadi/Applied-Regression-Analysis-in-R

R Basics

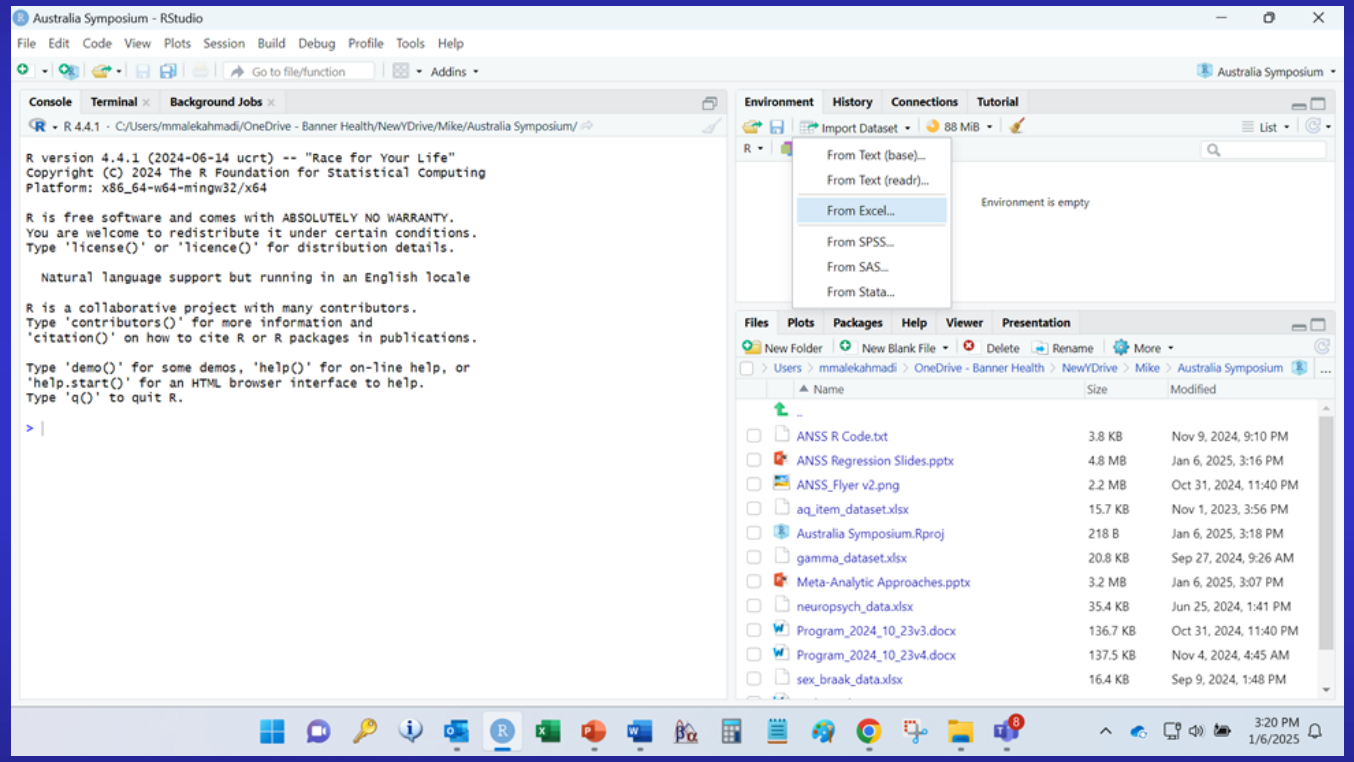
- R uses an object-based programming environment.
- Datasets and statistical models are objects within an R environment.
- R's programming language has a steep learning curve – Be Patient!

R Packages

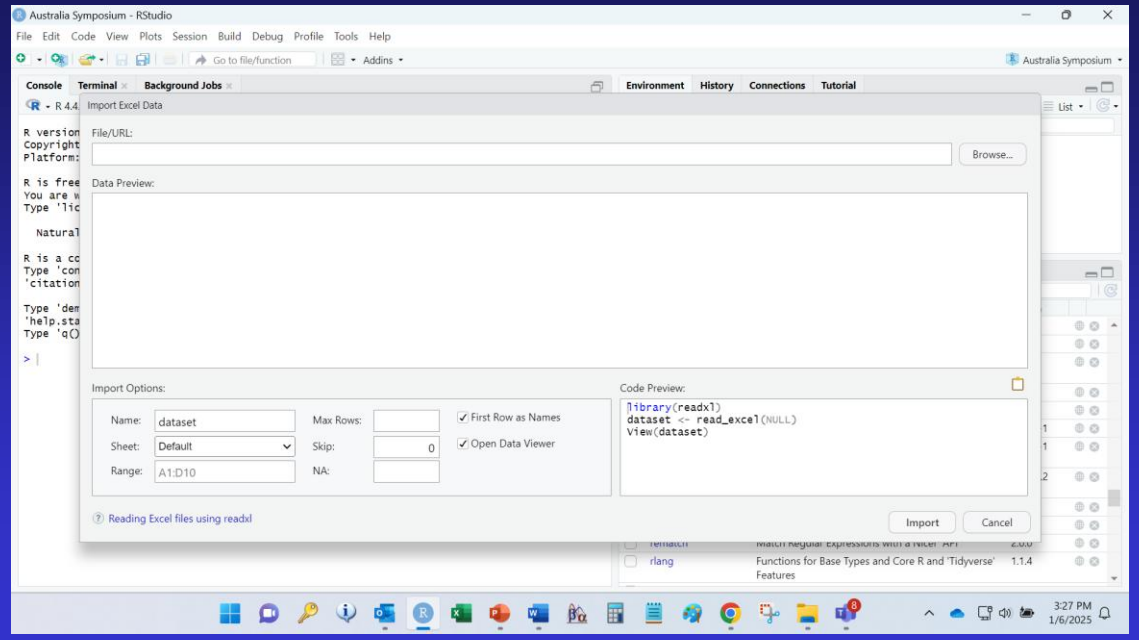
- MASS
- robust
- robustbase
- nlme
- ClinSigMeasures*

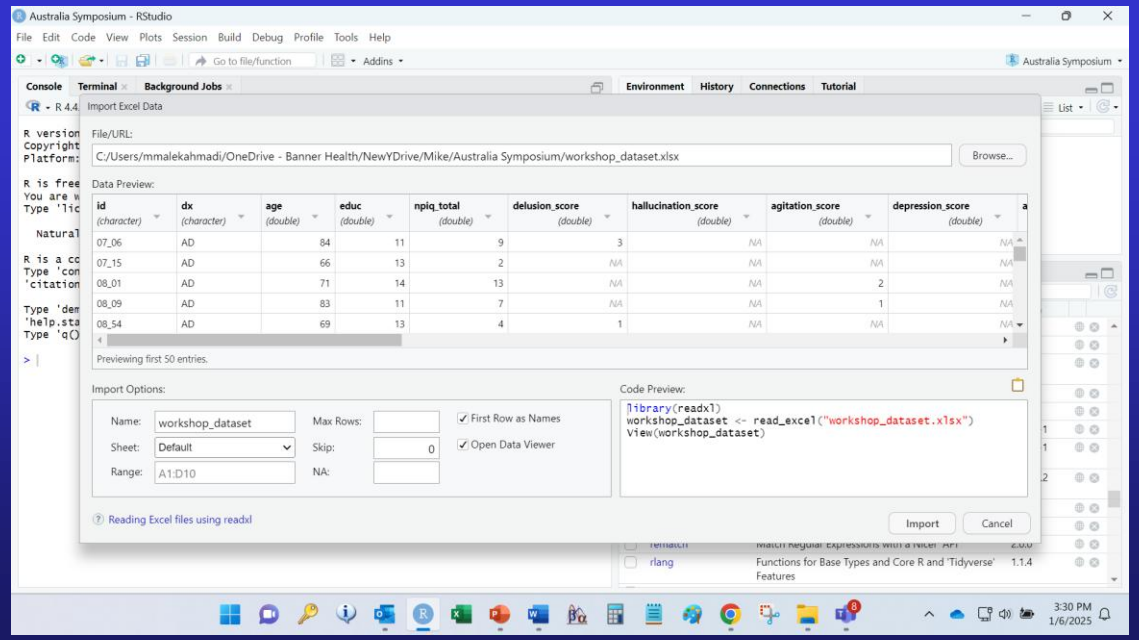
*Disclosure: I am the author and maintainer of the ClinSigMeasures package.

Importing Datasets into R Studio



- Click on “Import Dataset”.
- If prompted to install the “readxl” package, click Yes.
- Navigate to the folder on your computer where the dataset is saved.





- Select the file you want to import.
- You will then be shown a preview of the dataset with the first 50 lines.
- Select the file and click “Import”.

Australia Symposium - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Australia Symposium

workshop_dataset

Filter

	id	dx	age	educ	npiq_total	delusion_score	hallucination_score	agitation_score	depi
1	07_06	AD	84	11	9	3	NA	NA	
2	07_15	AD	66	13	2	NA	NA	NA	
3	08_01	AD	71	14	13	NA	NA	2	
4	08_09	AD	83	11	7	NA	NA	1	
5	08_54	AD	69	13	4	1	NA	NA	
6	09_04	MCI	82	19	1	NA	NA	NA	
7	09_23	MCI	79	13	1	NA	NA	NA	
8	09_27	AD	77	13	14	NA	2	2	
9	09_36	AD	67	13	9	1	3	1	

Showing 1 to 10 of 330 entries, 26 total columns

Console | Terminal | Background Jobs

R 4.4.1 · C:/Users/mmalekahmadi/OneDrive - Banner Health/NewYDrive/Mike/Australia Symposium/

```
  Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> library(readxl)
> workshop_dataset <- read_excel("workshop_dataset.xlsx")
> View(workshop_dataset)
```

Environment | History | Connections | Tutorial

Import Dataset | 130 MiB | List

R | Global Environment

Data

workshop_dataset | 330 obs. of 26 variables

Files | Plots | Packages | Help | Viewer | Presentation

Install | Update

	Name	Description	Version
☐	quantmod	Quantitative Financial Modelling Framework	0.4.26
☐	R6	Encapsulated Classes with Reference Semantics	2.5.1
☐	rappdirs	Application Directories: Determine Where to Save Data, Caches, and Logs	0.3.3
☐	rattle	Graphical User Interface for Data Science in R	5.5.1
☐	RColorBrewer	ColorBrewer Palettes	1.1-3
☐	Rcpp	Seamless R and C++ Integration	1.0.13-1
☐	RcppArmadillo	'Rcpp' Integration for the 'Armadillo' Templated Linear Algebra Library	14.2.2-1
☐	RcppEigen	'Rcpp' Integration for the 'Eigen' Templated Linear Algebra Library	0.3.4.0.2
☐	readr	Read Rectangular Text Data	2.1.5
☑	readxl	Read Excel Files	1.4.3
☐	rematch	Match Regular Expressions with a Nicer 'API'	2.0.0
☐	rlang	Functions for Base Types and Core R and 'Tidyverse' Features	1.1.4

3:34 PM
1/6/2025

Example of an R Object

- object_name<-function(variables, parameters, values)
- regression_model<-lm(dependent~variable1+variable2, data=dataset_name)
- summary(regression_model)

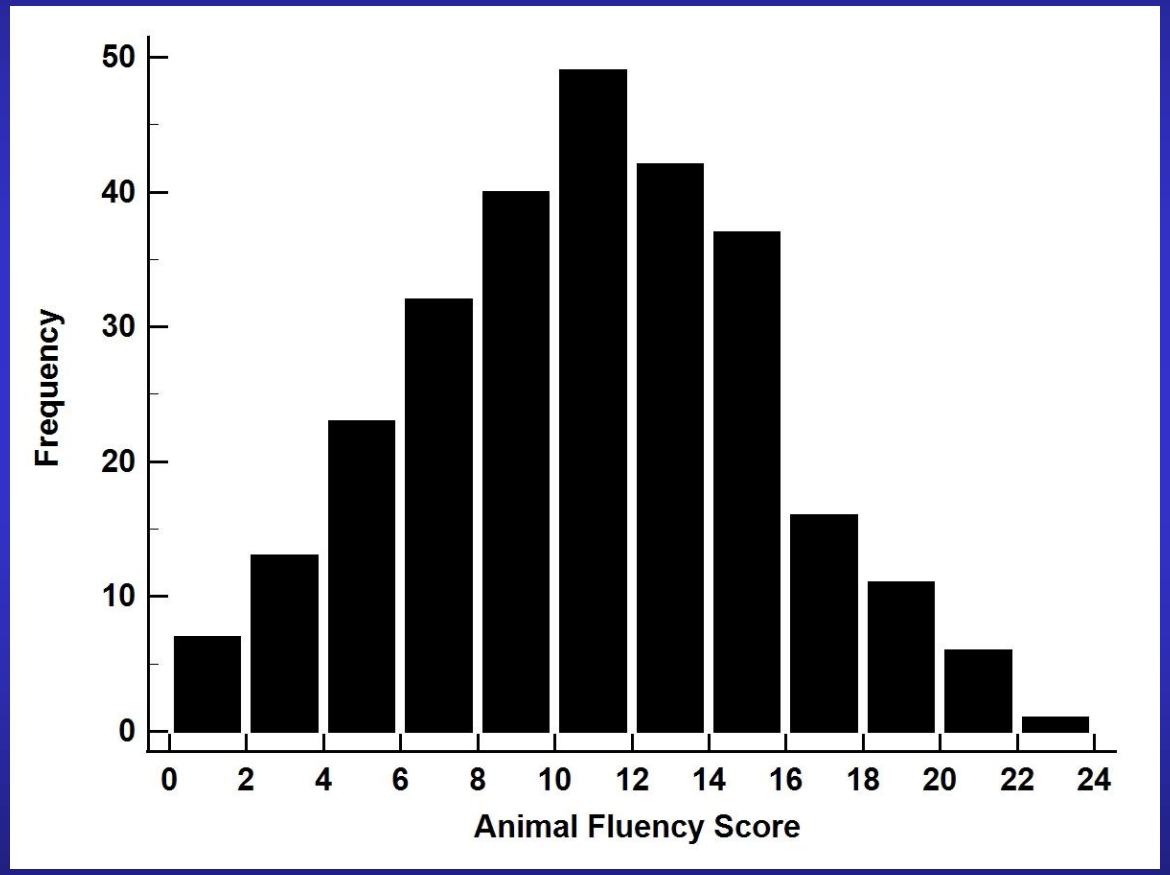
Distribution of Data

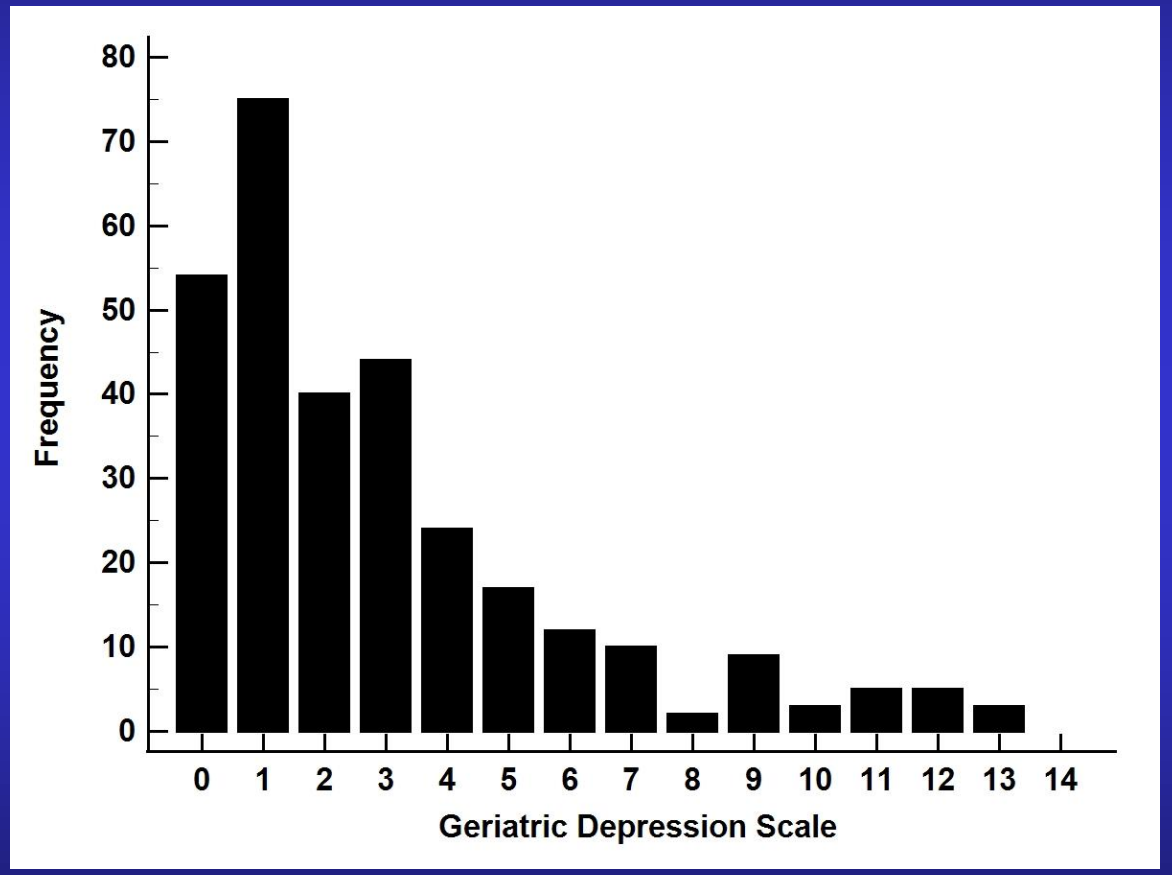
- One of the first steps in determining the distribution of a variable is to create histogram or frequency plot of the data.
- Q-Q plots are also helpful in assessing whether data satisfy the assumption of normality.
- There are also formal statistical tests that can help determine whether data are normally distributed.

Histograms

- One of the most common visualization methods to determine distribution shape.
- Very easy to determine whether data follow a normal distribution or whether data are significantly skewed.
- If significant skewness is present, histogram shape can help determine the proper analytic method.

Histograms

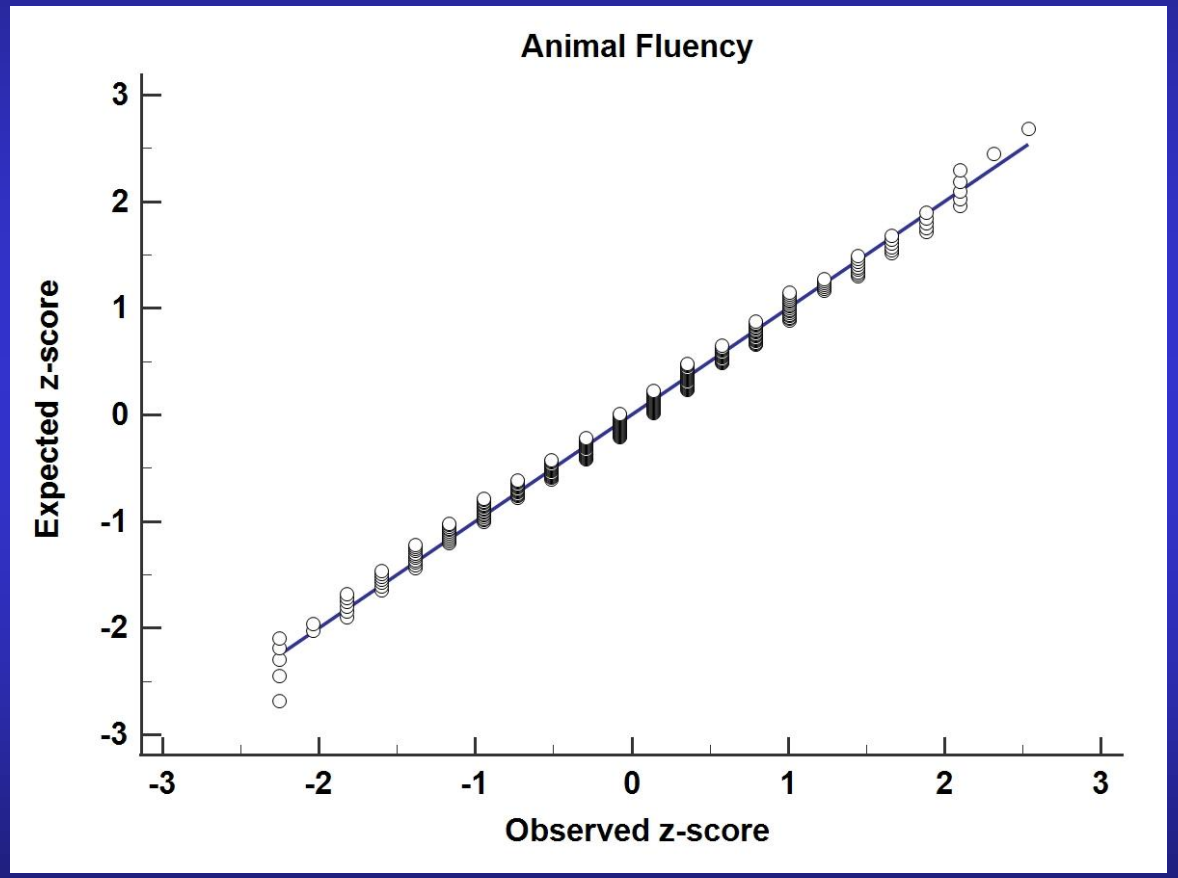

Frequency
0
10
20
30
40
50
0
2
4
6
8
10
12
14
16
18
20
22
24
Animal Fluency Score

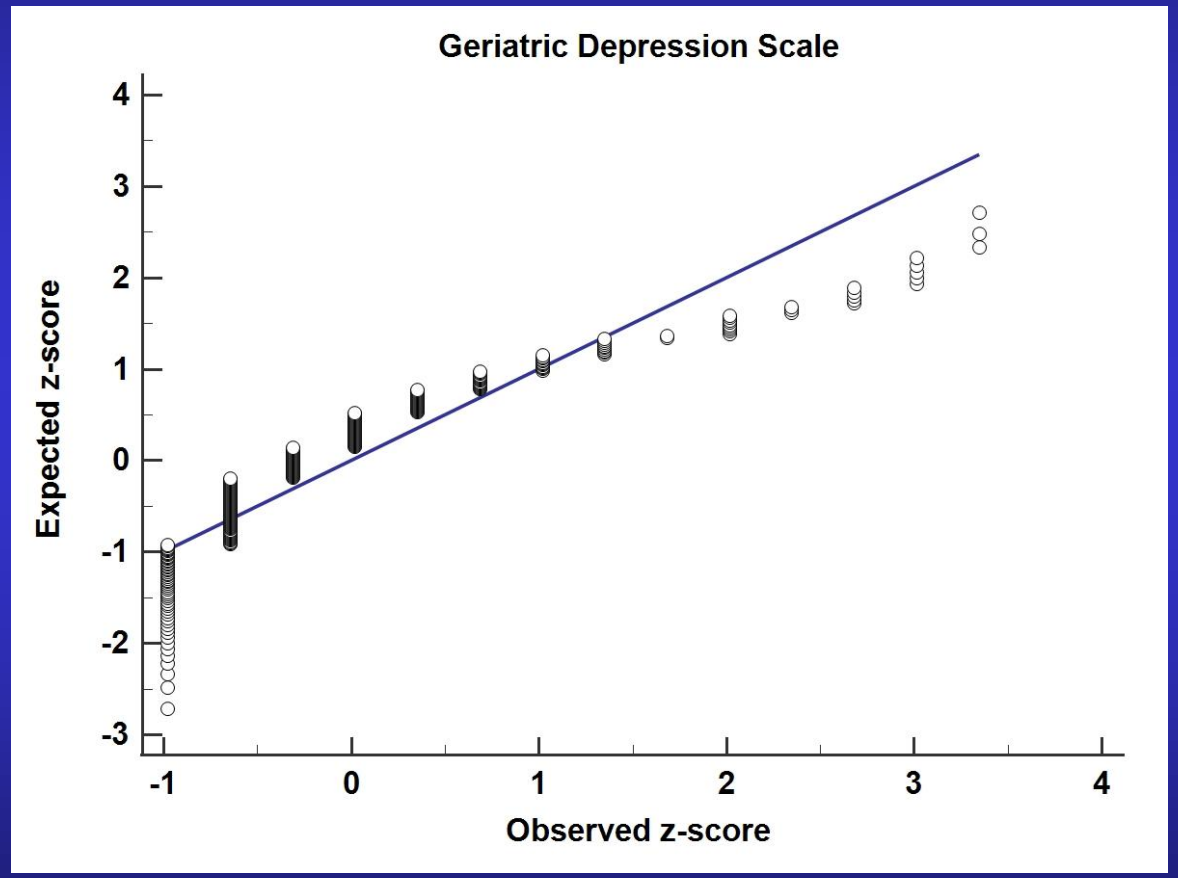

Frequency
0
10
20
30
40
50
60
70
80
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
Geriatric Depression Scale

Q-Q Plots

- Graphical approach to testing whether data satisfy the normality assumption.
- Data is plotted as quantiles from observed data and the expected distribution.
- 45° line is used to show where data should fall if normality assumption is met.
- Data points should generally follow along the 45° line, some dispersion of data is acceptable.

Q-Q Plots


Animal Fluency
Expected z-score
Observed z-score
3
2
1
0
-1
-2
-3
-3
-2
-1
0
1
2
3


Geriatric Depression Scale
Expected z-score
Observed z-score
4
3
2
1
0
-1
-2
-3
-1
0
1
2
3
4

Statistical Tests for Normality

- Most common statistical tests for normality are the Shapiro-Wilk and D'Agostino-Pearson tests.
- Both provide a p-value indicating whether a variable meets the assumption of normality.
- Important: higher p-values (>0.05) indicate that data are consistent with a normal distribution.

Statistical Tests for Normality

Variable	animal_fluency
Sample size	277
Lowest value	0.0000
Highest value	22.0000
Arithmetic mean	10.3574
95% CI for the Arithmetic mean	9.8141 to 10.9007
Median	10.0000
95% CI for the median	10.0000 to 11.0000
Variance	21.1001
Standard deviation	4.5935
Relative standard deviation	0.4435 (44.35%)
Standard error of the mean	0.2760
Coefficient of Skewness	-0.008199 (P=0.9547)
Coefficient of Kurtosis	-0.3775 (P=0.1370)
Shapiro-Wilk test for Normal distribution	W=0.9909 accept Normality (P=0.0830)

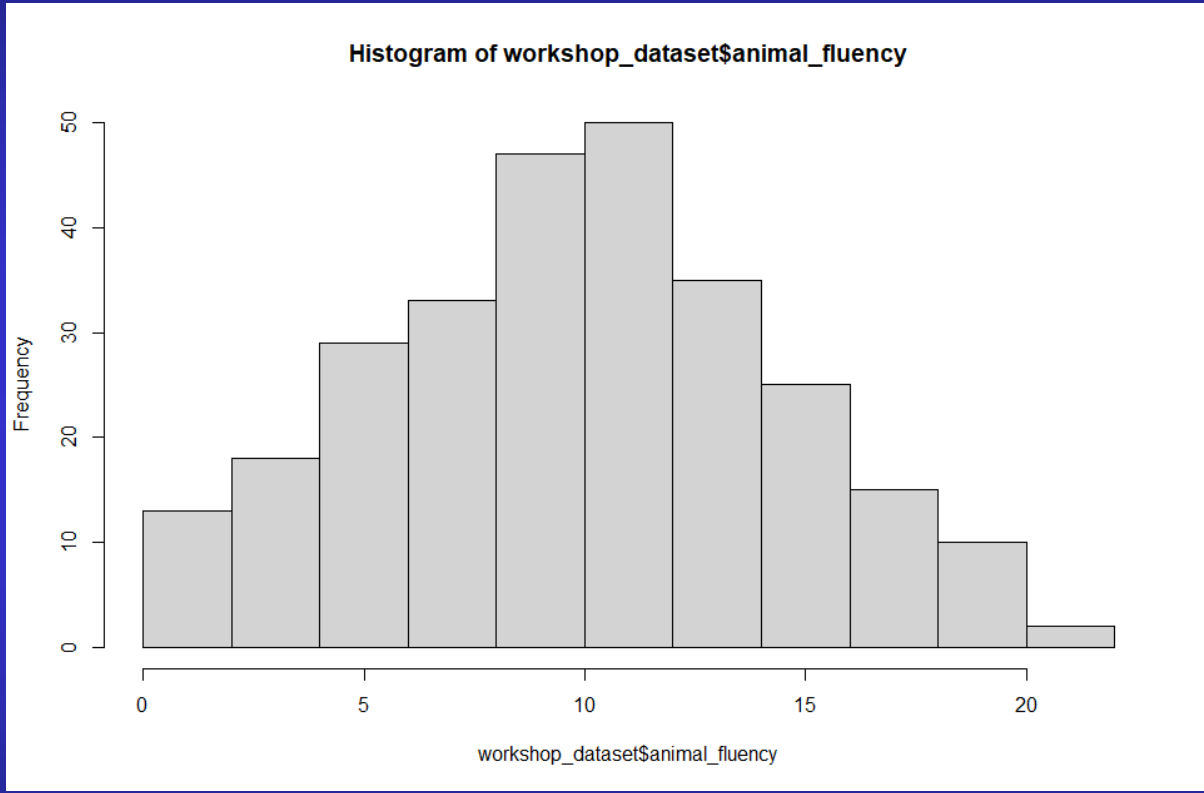
Variable	geri_dep_scale
Sample size	303
Lowest value	0.0000
Highest value	13.0000
Arithmetic mean	2.9406
95% CI for the Arithmetic mean	2.6006 to 3.2805
Median	2.0000
95% CI for the median	2.0000 to 2.0451
Variance	9.0428
Standard deviation	3.0071
Relative standard deviation	1.0226 (102.26%)
Standard error of the mean	0.1728
Coefficient of Skewness	1.4637 (P<0.0001)
Coefficient of Kurtosis	1.7459 (P=0.0001)
Shapiro-Wilk test for Normal distribution	W=0.8301 reject Normality (P<0.0001)

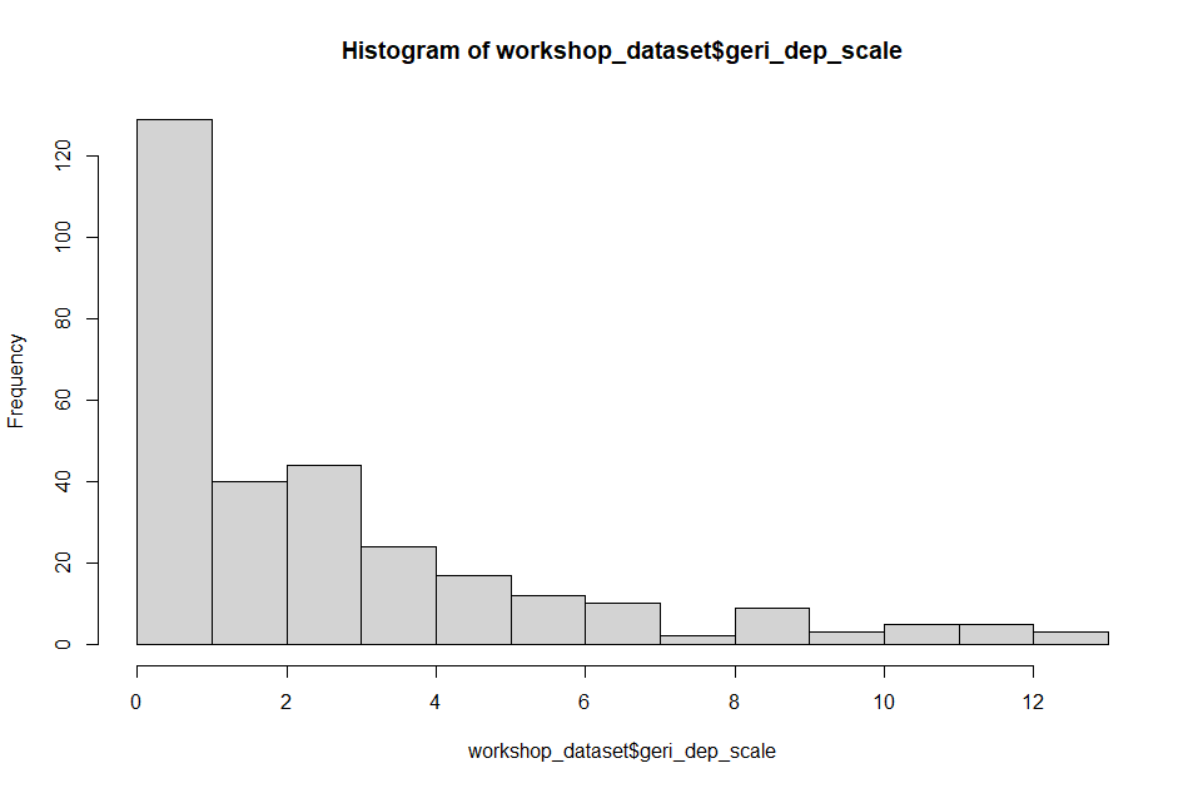
R Examples

- Histograms
- QQ Plots
- Statistical Tests for Normality

Histograms in R

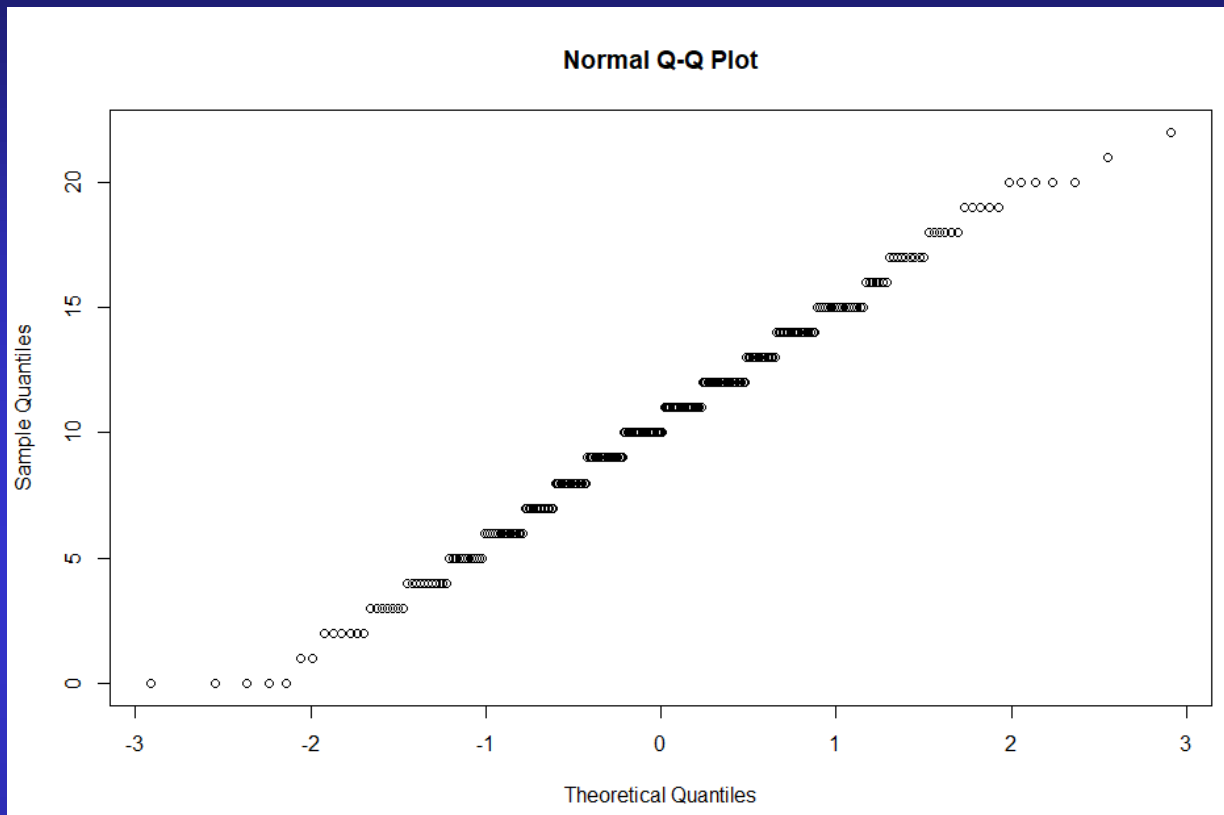
- Use the ‘hist’ command
- hist(datasetname$variable)
- hist(workshop_dataset$animal_fluency)
- hist(workshop_dataset$geri_dep_scale)

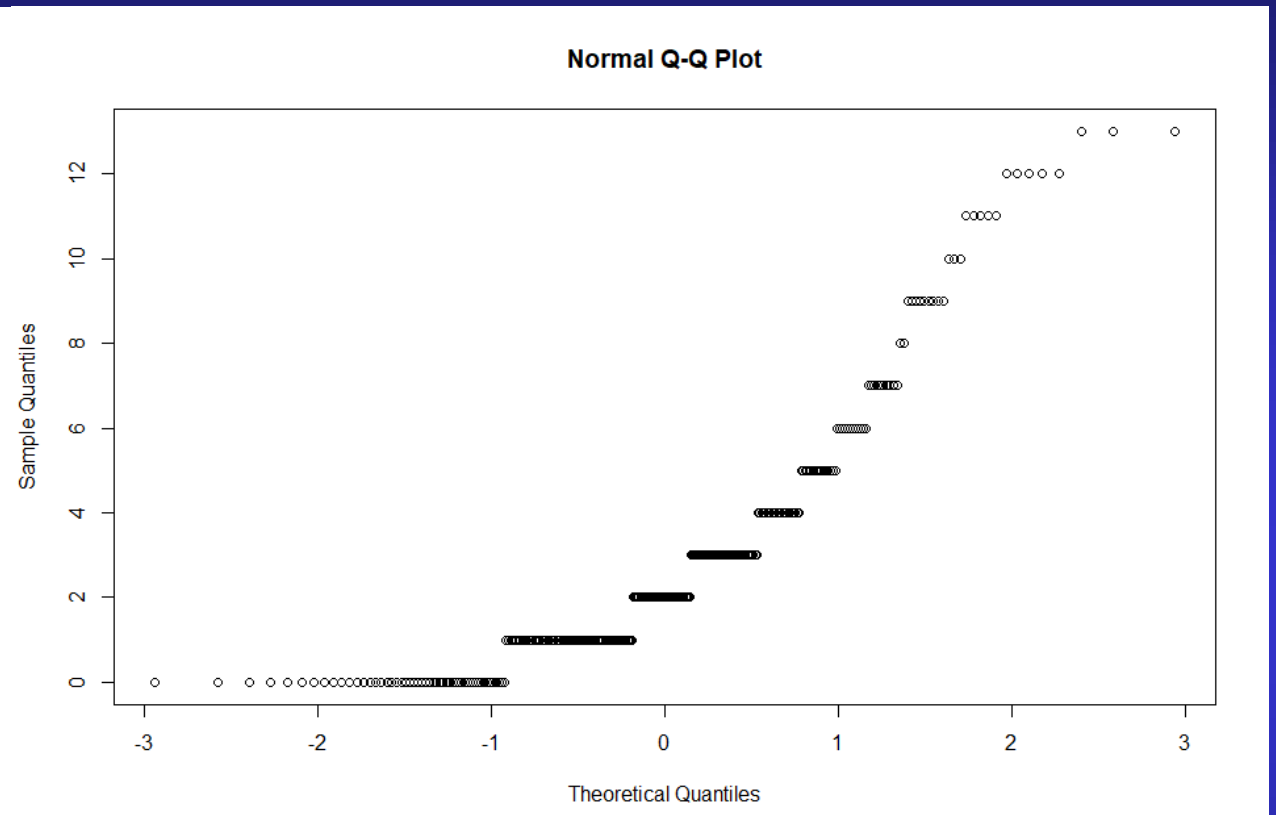

Histogram of workshop_dataset$animal_fluency
Frequency
0
10
20
30
40
50
0
5
10
15
20
workshop_dataset$animal_fluency


Histogram of workshop_dataset$geri_dep_scale
Frequency
0
20
40
60
80
100
120
0
2
4
6
8
10
12
workshop_dataset$geri_dep_scale

QQ Plots in R

- Use the ‘qqnorm’ command
- qqnorm(datasetname$variable)
- qqnorm(workshop_dataset$animal_fluency)
- qqnorm(workshop_dataset$geri_dep_scale)


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
0
5
10
15
20
-3
-2
-1
0
1
2
3


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles
0
2
4
6
8
10
12
-3
-2
-1
0
1
2
3

Shapiro-Wilk Test in R

- Use the ‘shapiro.test’ command
- shapiro.test(datasetname$variable)
- shapiro.test(workshop_dataset$animal_fluency)
- shapiro.test(workshop_dataset$geri_dep_scale)

Cautions of Shapiro-Wilk

- In large sample sizes the Shapiro-Wilk test may indicate a lack of normality when a visual inspection may indicate otherwise.

- Minor deviations from normality will not invalidate the results of a parametric test.

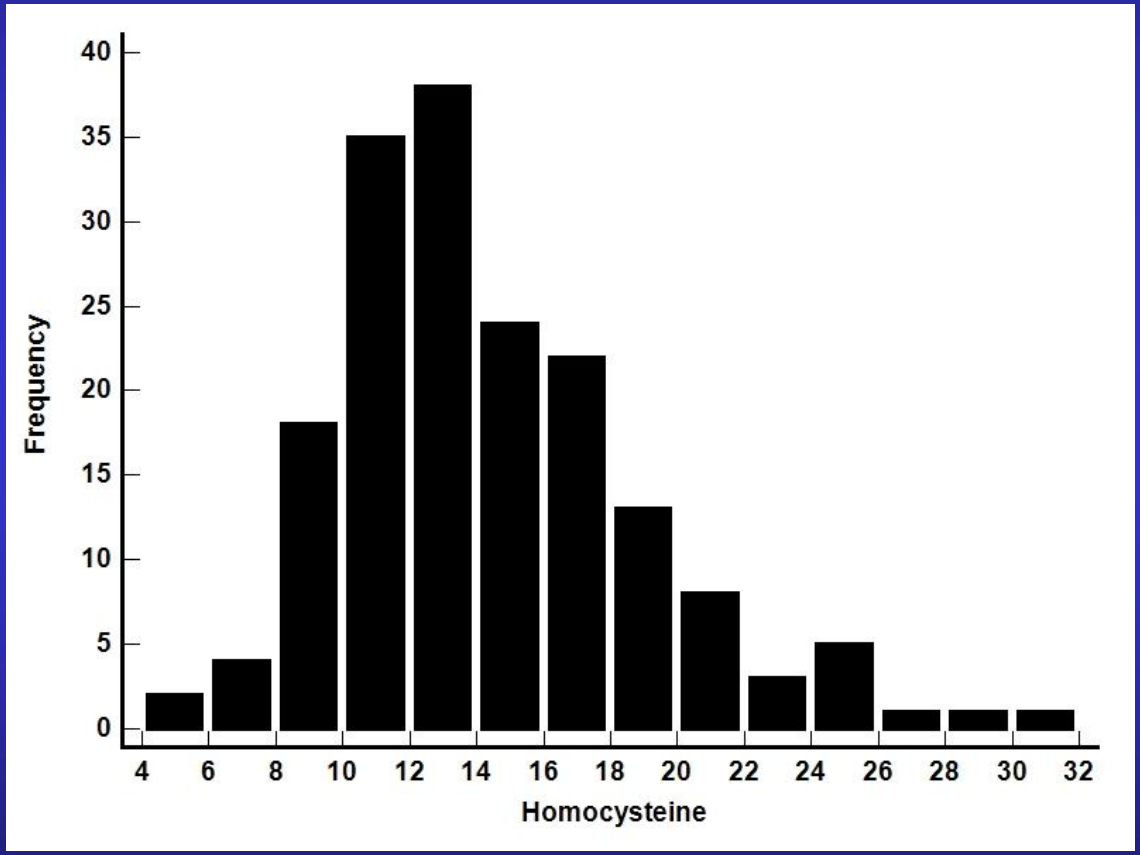
Log Transformation

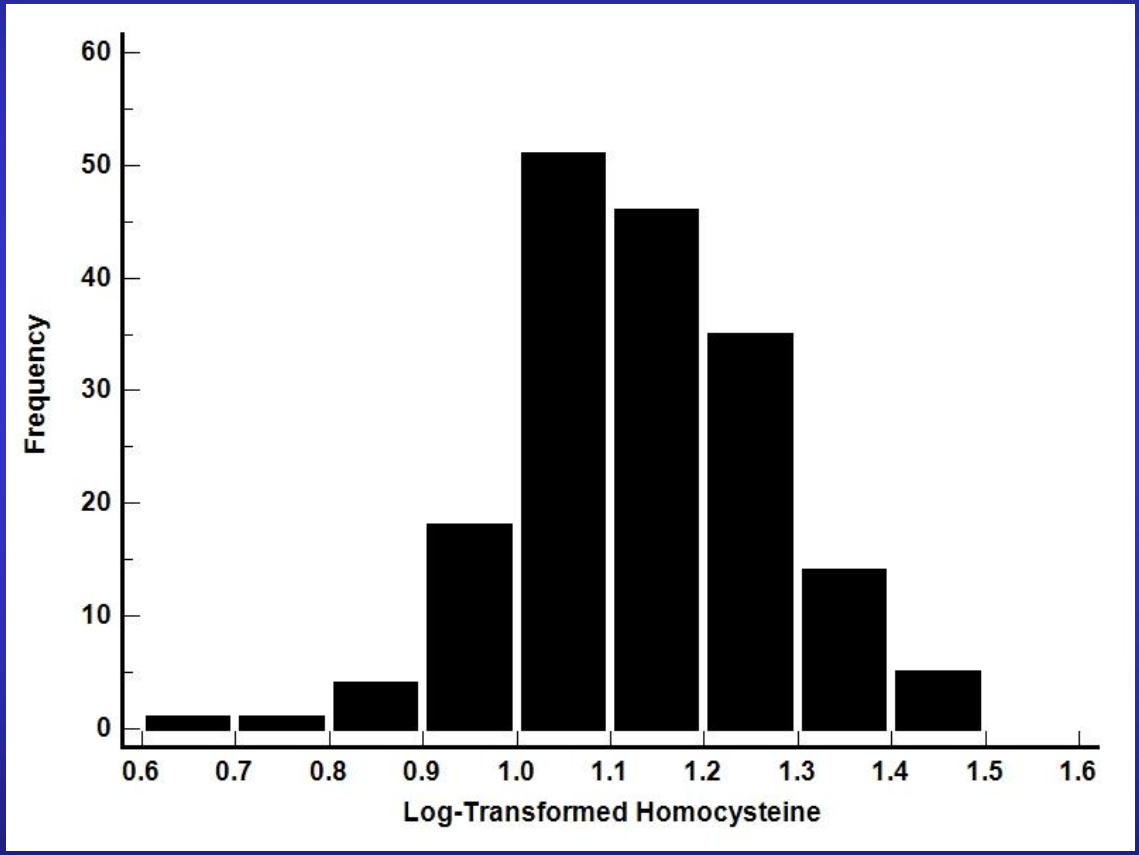
- A common approach to handling skewed data is to perform a log transformation which is intended to make the data fit a normal distribution.

- Despite its widespread, there are some precautions that need to be taken when log transformations are used.

1. Log transformations may not result in normally distributed data. Still need to perform visual inspection and normality tests of transformed data to ensure transformation was successful.
2. Log transformed values cannot be interpreted on the original scale of measurement.
3. If natural log (ln) transformation is used, mean and 95% CIs of ln-transformed values can be exponentiated. This gives the geometric mean and its 95% CI for the variable on the variable's original scale.

Log Transformation





Log Transformation

Variable	homocysteine
Sample size	175
Lowest value	4.0000
Highest value	30.5000
Arithmetic mean	14.2211
95% CI for the Arithmetic mean	13.5399 to 14.9024
Median	13.4000
95% CI for the median	12.5054 to 14.1946
Variance	20.8511
Standard deviation	4.5663
Relative standard deviation	0.3211 (32.11%)
Standard error of the mean	0.3452
Coefficient of Skewness	0.9010 (P<0.0001)
Coefficient of Kurtosis	0.9965 (P=0.0309)
Shapiro-Wilk test for Normal distribution	W=0.9513 reject Normality (P<0.0001)

Variable	log_hc
Sample size	175
Lowest value	0.6021
Highest value	1.4843
Arithmetic mean	1.1314
95% CI for the Arithmetic mean	1.1108 to 1.1520
Median	1.1271
95% CI for the median	1.0971 to 1.1521
Variance	0.01905
Standard deviation	0.1380
Relative standard deviation	0.1220 (12.20%)
Standard error of the mean	0.01043
Coefficient of Skewness	-0.1677 (P=0.3536)
Coefficient of Kurtosis	0.7920 (P=0.0650)
Shapiro-Wilk test for Normal distribution	W=0.9891 accept Normality (P=0.2008)

Linear Regression in R

- Use ‘lm’ function
- model_name<-lm(dependent_variable~variable1+variable2, data = datasetname)
- cdr_sob_animal<-lm(animal_fluency~cdr_sob, data=workshop_dataset)
- summary(cdr_sob_animal)

Linear Regression – Model Fit

- How do we determine whether the addition of a variable or variables improves the model?
- Model fit refers to how well the independent variables predict the dependent variable.
- For linear regression, the adjusted R^2 is the most common way to determine model fit.
- R^2 refers to the percent of variance that the independent variables account for in the dependent variable.
- Larger R^2 values indicate better model fit.

Linear Regression – Model Fit with Adjusted R^2

- Example using linear regression models

- Model 1: Linear regression model using the CDR Sum of Boxes as a predictor of Animal Fluency performance.
  - cdr_sob_animal<-lm(animal_fluency~cdr_sob, data=workshop_dataset)

- Model 2: Same model as above, but also includes the Geriatric Depression Scale, and MMSE scores.
  - cdr_sob_animal_adjusted<-lm(animal_fluency~geri_dep_scale+cdr_sob+mmse, data=workshop_dataset)

Linear Regression – Model Fit with Adjusted R^2

Model 1: Linear regression model using the CDR Sum of Boxes as a predictor of Animal Fluency performance.

- cdr_sob_animal<-lm(animal_fluency~cdr_sob, data=workshop_dataset)
- summary(cdr_sob_animal)

```
Call:
lm(formula = animal_fluency ~ cdr_sob, data = workshop_dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-9.3694 -3.2094 -0.3294  2.5806 11.3507

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.32944    0.32534  37.897   <2e-16 ***
cdr_sob     -0.48001    0.04893  -9.811   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.941 on 236 degrees of freedom
  (92 observations deleted due to missingness)
Multiple R-squared:  0.2897,    Adjusted R-squared:  0.2867
F-statistic: 96.25 on 1 and 236 DF,  p-value: < 2.2e-16
```

Linear Regression – Model Fit with Adjusted R^2

Model 2: Same model as above, but also includes Geriatric Depression Scale and MMSE as covariates.

- cdr_sob_animal_adjusted<-lm(animal_fluency ~geri_dep_scale+mmse+cdr_sob, data = workshop_dataset)
- summary(cdr_sob_animal_adjusted)

```
Call:
lm(formula = animal_fluency ~ geri_dep_scale + cdr_sob + mmse,
    data = workshop_dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-6.5738 -2.2759 -0.1231  1.8255  8.3182

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -0.56641    1.24258  -0.456    0.649
geri_dep_scale  0.02344    0.06734   0.348    0.728
cdr_sob        -0.06355    0.05573  -1.140    0.255
mmse            0.51034    0.04809  10.613   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.232 on 226 degrees of freedom
  (100 observations deleted due to missingness)
Multiple R-squared:  0.5089,    Adjusted R-squared:  0.5024
F-statistic: 78.06 on 3 and 226 DF,  p-value: < 2.2e-16
```

Linear Regression – Model Fit with Adjusted R^2

- In these two examples the adjusted R^2 value increased from 0.29 to 0.50 when the MMSE was added as a covariate in the model.

- While Model 2 appears to have better predictive value for Animal Fluency scores, we may question the results of this model given that the MMSE and CDR-SOB are highly correlated.

➢ cor(workshop_dataset$cdr_sob, workshop_dataset$mmse, method="spearman", use="complete.obs")

```
> cor(workshop_dataset$cdr_sob, workshop_dataset$mmse, method="spearman", use = "complete.obs")
[1] -0.7140827
```

- The presence of highly correlated predictor variables can be problematic which we will now discuss in further detail.

Covariate Selection in Regression Models

- Adjusting for covariates is a common practice when regression models are utilized.
- Selection of covariates is typically based on established biological/clinical relationships with the dependent variable.
- There are several factors to consider when selecting covariates.

Covariate Selection in Regression Models

- Multicollinearity – When covariates in a model are highly correlated this can reduce the precision of slope estimates for variables of interest.

- Stepwise Selection – Forward and Backward Selection procedures determine which covariates are included based on their p-values; may exclude covariates with biological/clinical significance.

- Subjects Per Covariate – A common rule of thumb is that there must be 10 subjects for each covariate included in a model. Covariate inclusion should be study-dependent and investigators should not rely strictly on this rule. Valid statistical models can be achieved with varying numbers of subjects per covariate.

Covariate Selection in Regression Models

- Tips for optimizing regression models:

- Select covariates based on biological/clinical significance
- Avoid the use of automated selection procedures
- Consider whether the covariates are highly correlated
- Do not over-fit a model with too many covariates
- Model fit can be assessed using Adjusted R^2 or Akaike's Information Criteria (AIC) value

Covariate Selection in Regression Models

- AIC values, by themselves, have no meaningful interpretation as they must be compared with one or more other AIC values.
- Lower AIC values indicate a better model fit, however there are no formal criteria for how much of a change in AIC values represents a substantially better model fit.
- One guideline suggests that a change of at least 2 AIC units is needed in order to state that a model has been improved, however others have suggested that a difference of at least 10 units indicates that one model can be favored over another.

Guidelines for Assessing AIC Differences

- Δ AIC <2: No substantial improvement
- Δ AIC 4<7: Modest/notable improvement
- Δ AIC >10: Substantial improvement

Behav Ecol Sociobiol 2011;65:13-21

Using AIC to Assess Model Fit

- Example using linear regression models

- Model 1: Linear regression model using the CDR Sum of Boxes as a predictor of Animal Fluency performance.
  - cdr_sob_animal<-lm(animal_fluency~cdr_sob, data=workshop_dataset)

- Model 2: Same model as above, but also includes Geriatric Depression Scale and MMSE as covariates.
  - cdr_sob_animal_adjusted<-lm(animal_fluency ~geri_dep_scale+mmse+cdr_sob, data = workshop_dataset)

Using AIC to Assess Model Fit

	AIC Value
Model 1	1332
Model 2	1198

- The difference in AIC values is 134 which favors Model 2
- Model 2 provides a better estimate of the association between CDR Sum of Boxes and Animal Fluency performance.

```
> AIC(cdr_sob_animal)
[1] 1332.209
> AIC(cdr_sob_animal_adjusted)
[1] 1198.329
```

Alternatives to Linear Regression

- Robust Regression
- Negative Binomial Regression
- GLM Gamma Regression

Robust Regression

- In cases where the dependent variable does not meet the assumption of normality other types of regression analysis can be used.
- Robust regression is an accessible and intuitive option.
- The median is used as the regressor, whereas linear regression uses the mean.

Robust Regression Example

- In R, use ‘robust’ and ‘robustbase’ packages.
- Using the ‘lmRob’ function, we will utilize the same structure as the linear model to create the robust model.
- For this example we will use the animal_fluency and faq variables in “workshop_dataset”.

Robust Regression Example

- We will use scores from the Functional Activities Questionnaire (faq) as the dependent variable.

- Examine the distribution

hist(workshop_dataset$faq)

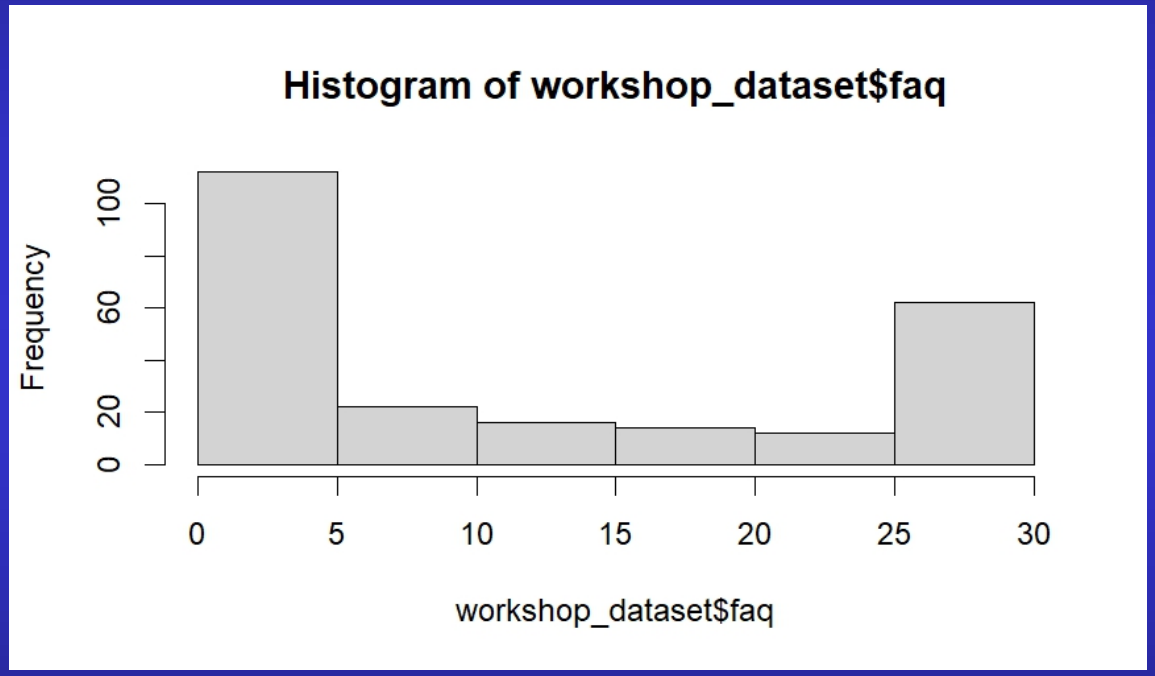
qqnorm(workshop_dataset$faq)

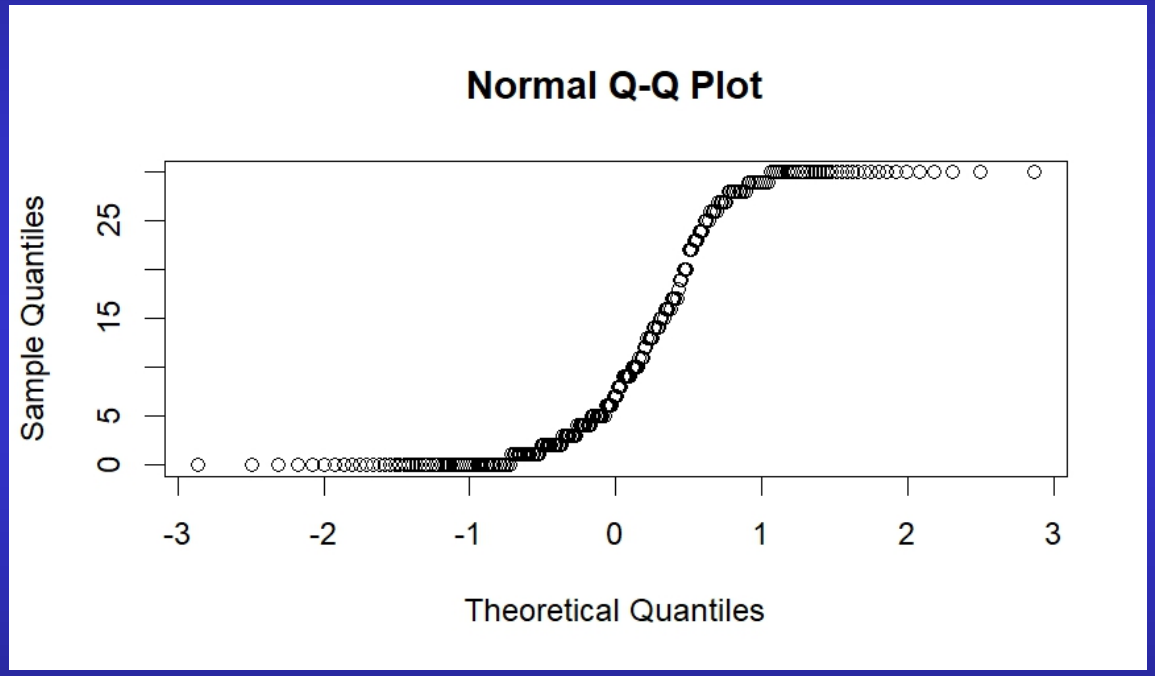
- Create linear and robust regression models using the Animal Fluency score (animal_fluency) as the predictor variable.

➢ linear_animal_fluency_faq<-lm(faq~animal_fluency, data=workshop_dataset)

➢ robust_ animal_fluency_faq <-lmRob(faq~animal_fluency, data=workshop_dataset)

Robust Regression Example





Robust Regression Example

```
Call:
lm(formula = faq ~ animal_fluency, data = workshop_dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-15.077  -8.191  -1.922   7.905  24.348

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    23.4241     1.6324  14.350  < 2e-16 ***
animal_fluency -1.2694     0.1464  -8.671 1.58e-15 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.597 on 197 degrees of freedom
  (131 observations deleted due to missingness)
Multiple R-squared:  0.2762,    Adjusted R-squared:  0.2725
F-statistic: 75.18 on 1 and 197 DF,  p-value: 1.581e-15
```

```
Call:
lmRob(formula = faq ~ animal_fluency, data = workshop_dataset)

Residuals:
    Min      1Q  Median      3Q     Max
-7.1452 -3.8414  0.3841 13.5017 26.0802

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    10.3707     2.0109   5.157 6.08e-07 ***
animal_fluency -0.4608     0.1649  -2.794  0.00571 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

On the left is the linear model and on the right is the robust model. Although both models indicate a statistically significant association, the linear model yields a much larger regression coefficient (-1.2694) compared to the robust model (-0.4608). Given the distribution of the FAQ, it would be best to report the robust regression results.

Caveats to Robust Regression

- Although robust regression can be used with many different distributions of data, there are some limitations.
- It is difficult to assess model fit as robust regression models do not yield Adjusted R^2 or AIC values.
- Can use Multiple R^2 to compare between other models, but this value may not be reliable for models with several covariates.

Poisson and Negative Binomial Models

- Poisson models have the following assumptions:

1. Dependent variable is count data (integers).
2. Mean and variance of the dependent variable are equal or approximately equal.

- NB models still require count data to be used, but can be used when variance is greater than the mean.

NPI-Q Total Score Example

- Proposed analysis: CDR Sum of Boxes as a predictor of NPI-Q Total Score.

- Generate histogram and QQ-plot for NPI-Q Total Score

```
hist(workshop_dataset$npiq_total)
qqnorm(workshop_dataset$npiq_total)
```

NPI-Q Total Score Example

- Since the dataset includes MCI and AD cases, analyzing the groups separately may help us understand the distribution better.

- To create separate datasets for MCI and AD:

➢ mci_dataset<-subset(workshop_dataset, dx=="MCI")

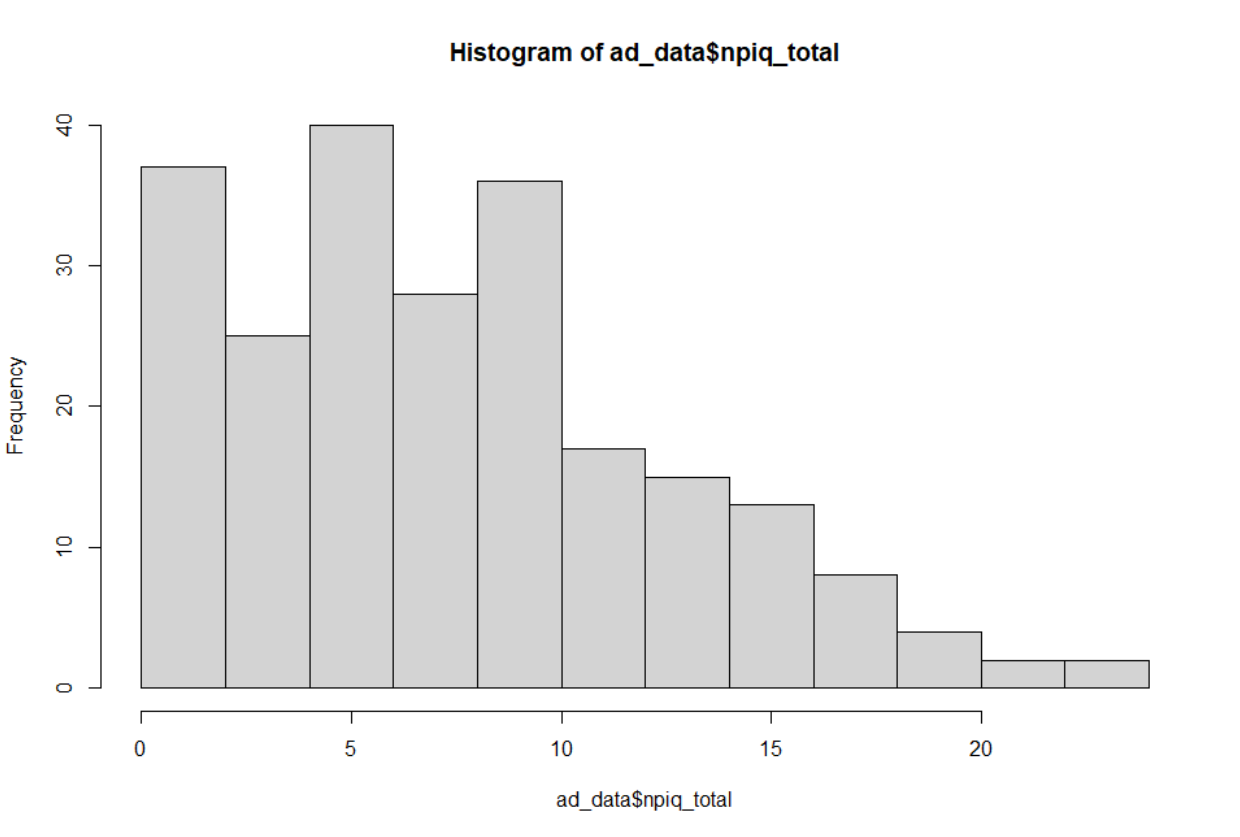
➢ ad_dataset<-subset(workshop_dataset, dx=="AD")

NPI-Q Total Score Example

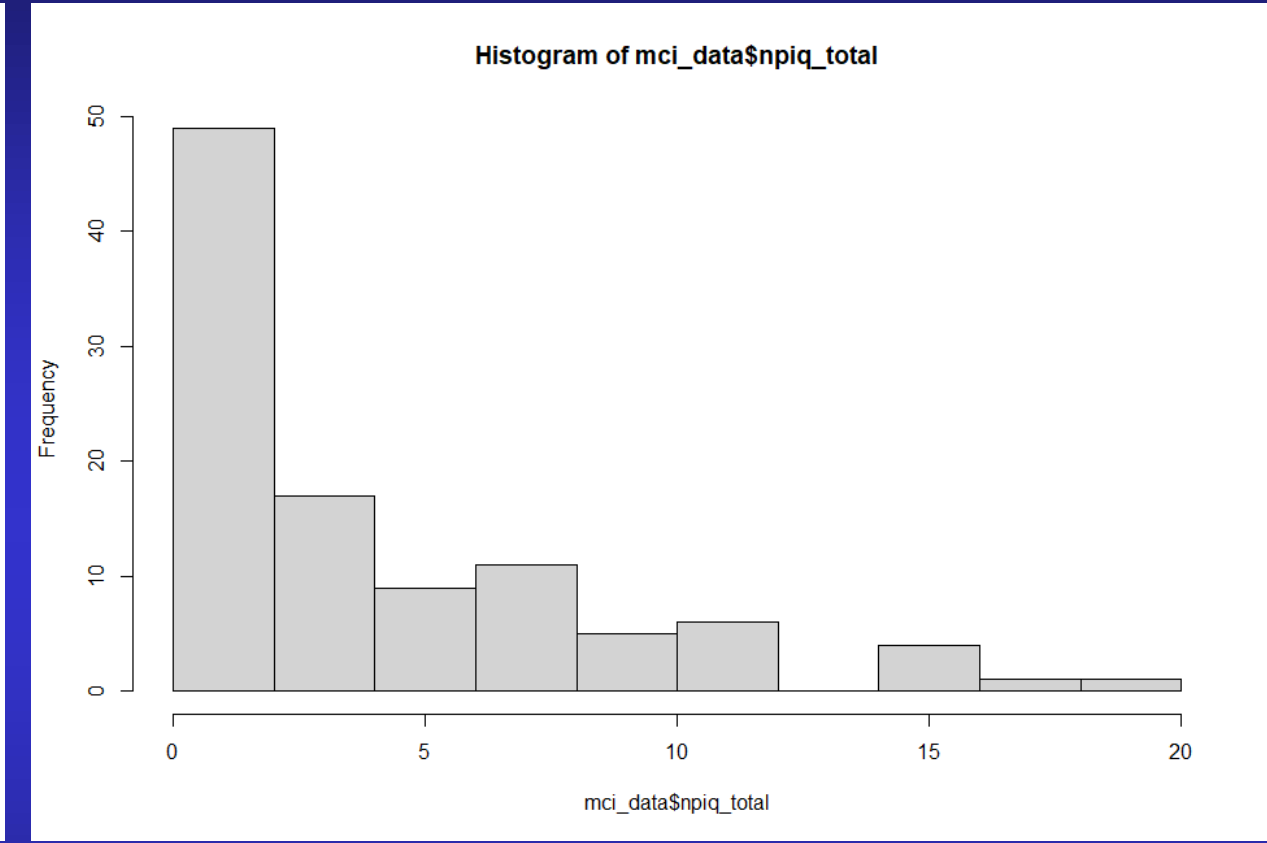
- Now we can examine the histograms for the NPI-Q Total Score in each group.

- hist(mci_dataset$npiq_total)
- hist(ad_dataset$npiq_total)

AD Group



MCI Group



Both distributions are significantly right-skewed, but the histogram for the MCI group indicates a very high frequency for a score of zero. We will now apply Poisson and Negative Binomial models to each dataset to determine which is best.

R Code for Poisson and Negative Binomial Models

- For all models, we will be using Mini-Mental State Exam (MMSE) scores to predict NPI-Q Total Scores in the AD group.

- Poisson Model

➢ poisson_ad<-glm(npiq_total~mmse, family=poisson(link="log"), data=ad_dataset)

- Negative Binomial Model

➢ library(MASS)

➢ negative_binomial_ad<-glm.nb(npiq_total~mmse, data=ad_dataset)

Comparison of AD Group Models

Poisson Model

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.33623    0.05552  42.078  < 2e-16 ***
mmse        -0.01789    0.00308  -5.807 6.37e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 822.09  on 210  degrees of freedom
Residual deviance: 788.94  on 209  degrees of freedom
  (16 observations deleted due to missingness)
AIC: 1527.4
```

Negative Binomial Model

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.366081   0.112037  21.119  < 2e-16 ***
mmse        -0.019676   0.005988  -3.286  0.00119 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(2.4619) family taken to be 0.8763562)

    Null deviance: 259.86  on 210  degrees of freedom
Residual deviance: 251.15  on 209  degrees of freedom
  (16 observations deleted due to missingness)
AIC: 1267.6
```

1. The regression coefficients (Estimate) are the same for both models and are statistically significant.
2. The standard error (Std. Error) for the coefficient of the NB model is twice that of the Poisson model.
3. The residual deviance for the Poisson model (788) is more than 3X that of the NB model (251).
4. AIC value for NB model is substantially lower (1267) than the Poisson AIC (1527).
5. Indications are that the NB model provides a more robust estimate of the association between MMSE and NPI-Q Total Score in AD subjects.

Additional Examination of NB Models

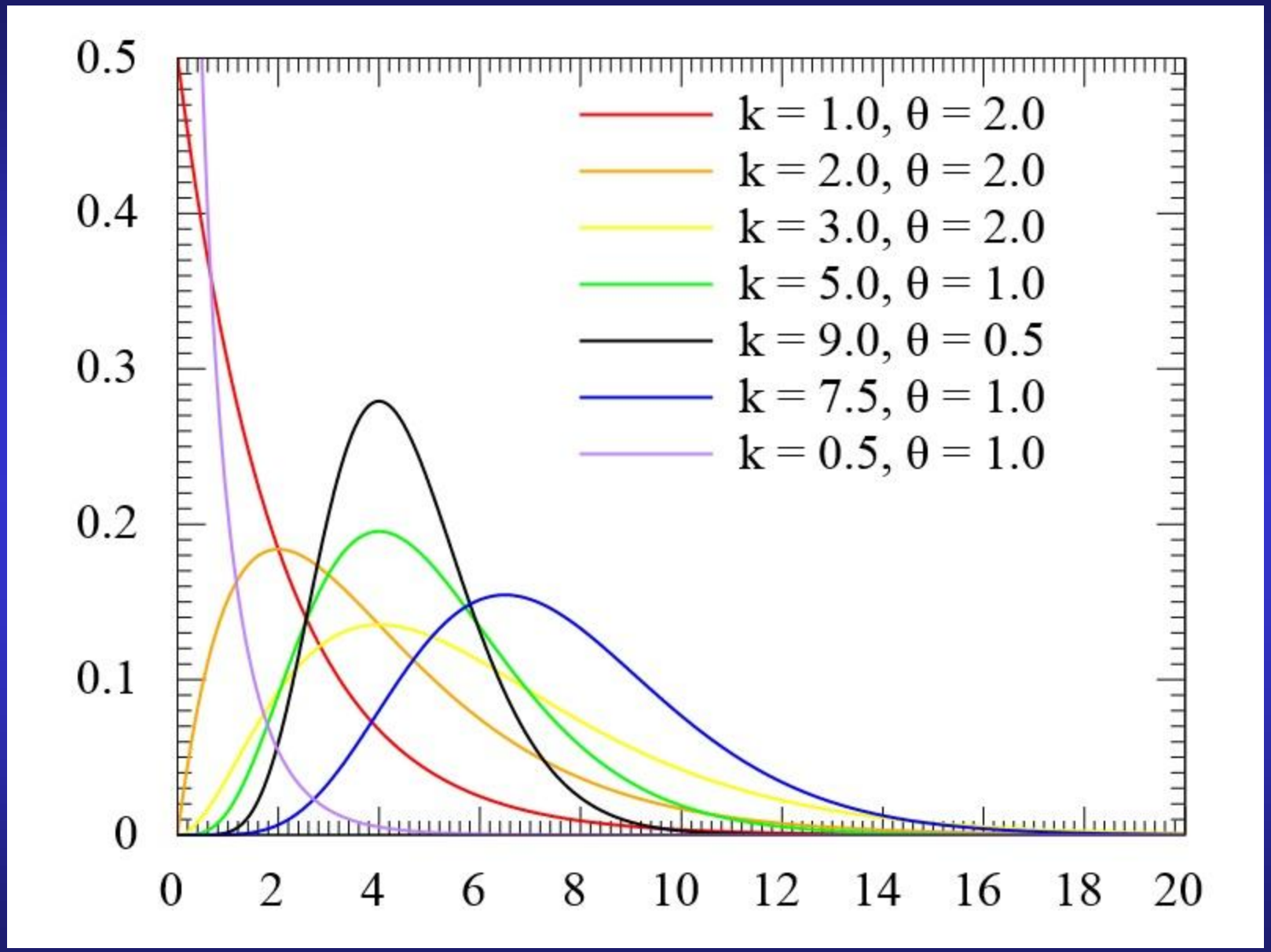
- Although the difference in AIC and Residual Deviance values favor the NB models we can obtain a more objective assessment of whether the excess variance is properly accounted for.

- The residual deviance values can be tested in the Chi-square distribution using their corresponding degrees of freedom that is in the model output.

- In R, this can be done using the following:
  - 1-pchisq(residual deviance value, degrees of freedom)
  - The resulting p-value will indicate if the model is ($p>0.05$) or is not ($p<0.05$) a good fit for the data.

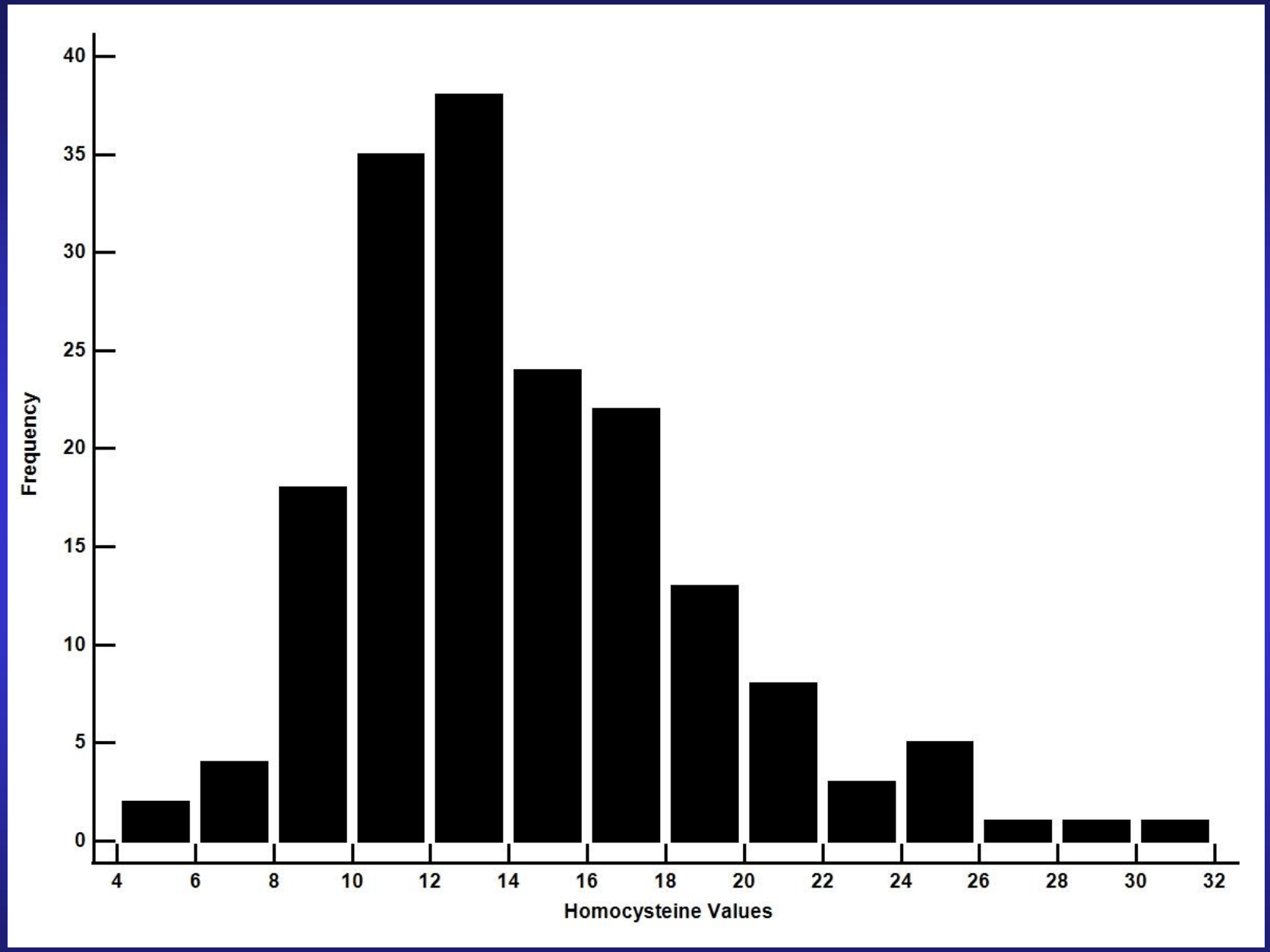
The Gamma Distribution

- What if the outcome variable is not an integer?
- Under the generalized linear model (GLM) family, the gamma distribution can be specified.
- However, the dependent variable cannot have zeroes or negative numbers.


0.5
0.4
0.3
0.2
0.1
0
0
2
4
6
8
10
12
14
16
18
20
k = 1.0, θ = 2.0
k = 2.0, θ = 2.0
k = 3.0, θ = 2.0
k = 5.0, θ = 1.0
k = 9.0, θ = 0.5
k = 7.5, θ = 1.0
k = 0.5, θ = 1.0

Gamma GLM Example

- In this example, we will use homocysteine as the dependent variable.
- Age will be used as the independent variable.
- Linear and GLM gamma models will be compared on the regression coefficients for age as well as AIC values.



hist(gamma_dataset$homocysteine)

Gamma GLM Example

- age_hc_linear<-lm(homocysteine~age, data=gamma_dataset)
- summary(age_hc_linear)
- AIC(age_hc_linear) *because AIC is not in output of lm models.
- age_hc_gamma<-glm(homocysteine~age, family=Gamma(link="log"), data=gamma_dataset)
- summary(age_hc_gamma)

Gamma GLM Example

	Coeff.	SE	P-value	AIC Value
Linear Model (Age)	0.09	0.04	0.02	855
Gamma Model (Age)	0.006	0.003	0.02	829

- Gamma model is favored due to lower AIC value (829) relative to the linear model AIC (855).
- The linear model may be over-estimating the association between age and homocysteine level.

Break

Logistic Regression

- Logistic regression falls under the GLM family and is used when the outcome variable is dichotomous.
- Like other types of regression, the independent variables can be continuous or categorical.
- Regression coefficients are converted into odds ratios (OR) for proper interpretation.

Odds Ratio

- Odds Ratio (OR) is the statistic of interest derived from logistic regression.
- OR is a measure of effect size used when the outcome variable is dichotomous (yes/no, positive/negative, etc.)
- Interpreted in terms of percent risk increase or decrease.

Odds Ratio

- OR interpretation is based on the value of 1.00 which indicates no risk increase or decrease.
- OR values greater than 1.00 indicate risk increase. OR values less than 1.00 indicate risk decrease.
- OR = 1.23: 23% more likely to have X given Y.
- OR = 3.50: 3.5 times more likely to have X given Y.
- OR = 0.89: 11% less likely to have X given Y.

Odds Ratio

	Disease Positive	Disease Negative
Risk Factor Present	A	B
Risk Factor Absent	C	D

OR = (A*D) / (B*C)

Odds Ratio – Sex and Braak Stage

	Braak 0 – II	**Braak III – V**
Male	31	32
Female	14	46

Odds Ratio	3.18
95% CI	1.47 to 6.91
Significance level	P = 0.003

Mufson et al. *Neurobiol Aging* 2016;37:147-153.

Odds Ratio

- When interpreting an OR for a continuous variable we interpret the risk increase, or decrease, relative to the unit of measure for the predictor variable.

- OR = 1.06; for every unit increase in X, the risk of Y increases by 6%.

- OR = 0.89; for every unit increase in X, the risk of Y decreases by 11%.

Logistic Regression Example in R

- Be sure that the outcome variable is dichotomous and coded with 0 and 1 (0 = control, 1 = case).
- Numeric or categorical variables can be used as predictor variables.
- Will also need to obtain confidence intervals for the OR estimate (derived from regression coefficient in output).

Logistic Regression Example in R

- In the “sex_braak_data” dataset the binary outcome variable is “braak_01” where 0 = Low Braak Stage and 1 = High Braak Stage.

- Data for sex (sex_01) is 0 = Male, 1 = Female).

- braak_sex_logistic<-glm(braak_01~sex_01, family=binomial(link="logit"), data= sex_braak_data)

Logistic Regression Example in R

```
Call:
glm(formula = braak_01 ~ sex_01, family = binomial(link = "logit"),
    data = sex_braak_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.03175    0.25201   0.126  0.89975
sex_01       1.15784    0.39582   2.925  0.00344 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 161.55  on 122  degrees of freedom
Residual deviance: 152.51  on 121  degrees of freedom
AIC: 156.51

Number of Fisher Scoring iterations: 4
```

```
> confint.default(braak_sex_logistic)
                 2.5 %    97.5 %
(Intercept) -0.4621780 0.5256754
sex_01       0.3820371 1.9336336
```

Logistic Regression Example in R

- The regression coefficient for sex is 1.158 and its 95% confidence interval is (0.382, 1.934).
- Exponentiate the regression coefficient and confidence interval values to obtain OR with its 95% confidence intervals.
- Exp(1.158) = 3.18; (Exp(0.382) = 1.47, Exp(1.934)= 6.91)
- OR = 3.18 (1.47, 6.91)
- Females are 3.18 times as likely to have higher Braak stage than males.

Linear Mixed-Effects Models

- For longitudinal data, linear mixed-effects models (LMMs) are a popular choice.
- Still retains the assumption of normality for the dependent variable, but…
- User must also consider the correlation/covariance matrix of correlated time-series data.

Linear Mixed-Effects Models

- The structure of LMMs is similar to linear regression models that use cross-sectional data with an independent variable of interest and other selected covariates.
- The inclusion of the random effect is what differentiates LMMs from typical linear regression models.
- Random effect refers to specific grouping or classification within a dataset that introduces error into the estimates of regression model.

Subject-Level Data as Random Effects

- Although random effects can take different forms depending on the context of the data, the most common application of random effects is to account for between-subject variation on both baseline values and trajectories of change.

- This approach also allows us to make estimates of change at the subject level, but the important point is that the LMM estimates will have accounted for differences between subjects.

Subject-Level Data as Random Effects

- In order to account for between-subject differences in longitudinal data, the column containing subject ids is treated as a categorical variable where each id has two or more observations.

Subject_id	Time	Sex	Outcome
455	1	M	34
455	2	M	27
455	3	M	22
873	1	F	44
873	2	F	39
873	3	F	38

LMM Covariance Structure

- An important component of LMMs is how the covariance structure is specified.
- The covariance structure refers to how the data for the dependent variable are correlated over time (e.g., increase, decrease, constant, unknown, etc.).
- Specifying a covariance structure within the LMM model can enhance the robustness of the model.

Common Covariance Structures for LMMs

- Variance Components (VC) – the most common structure and is often set as the default structure in many statistical packages; assumes a constant correlation across measurements

- Auto-regressive order 1 (AR(1)) – very useful for longitudinal data as it assumes that the correlation between measurements decreases as they become farther apart (i.e., dependent variable reliably increases or decreases over time).

LMM Example in R

- Using the ‘nlme’ package, we will use the ‘lme’ function.

```
model_name<-lme(cognition~time+group+time*group, random=~1+time|subject_id, correlation=corAR1, data=dataset)
```

- The estimate for time-based change of the independent (group) and dependent variable (cognition) is generated from the time*group interaction term in the model.

- The random effect (1+time|subject_id) is the intercept and slope (baseline score and rate of change) for each subject and helps minimize the impact of between subject differences on the model’s estimates.

LMM Example in R

```
moca_longitudinal_sex_diff<-
lme(moca~time+sex+time*sex,
random=~1+time|study_id, data=lmm_example)

summary(moca_longitudinal_sex_diff)
```

```
Linear mixed-effects model fit by REML
  Data: lmm_example
       AIC     BIC    logLik
  7375.403 7418.4 -3679.701

Random effects:
 Formula: ~1 + time | study_id
 Structure: General positive-definite, Log-Cholesky parametrization
            StdDev    Corr
(Intercept) 1.4830687 (Intr)
time        0.2912888 0.364
Residual    1.9603192

Fixed effects:  moca ~ time + sex + time * sex
                 Value  Std.Error   DF   t-value p-value
(Intercept)  26.407088 0.15378220 1130 171.71745  0.0000
time         -0.165951 0.05379885 1130  -3.08465  0.0021
sexMale      -0.251040 0.29232879  465  -0.85876  0.3909
time:sexMale -0.131494 0.10235422 1130  -1.28470  0.1992
 Correlation:
             (Intr) time   sexMal
time         -0.661
sexMale      -0.526  0.348
time:sexMale  0.348 -0.526 -0.662

Standardized Within-Group Residuals:
        Min          Q1         Med          Q3         Max
-2.65821492 -0.57307473  0.05072696  0.64693058  2.58133570

Number of Observations: 1599
Number of Groups: 467
```

Broad Areas of Statistical Errors

- Incorrect statistical test used to address a research question.
- Incorrect interpretation of statistical tests.
- Incorrect use of statistical terminology.

Which One Do I Use?

- Standard Deviation vs. Standard Error
- Multivariate vs. Multivariable
- Mediation vs. Moderation

Standard Deviation vs. Standard Error

- One of the most common errors is to report the standard error as a measure of variability instead of the standard deviation.

- Standard error: Provides an estimate of precision when comparing a sample to a population.

- Standard deviation: A measure of variability for a sample.

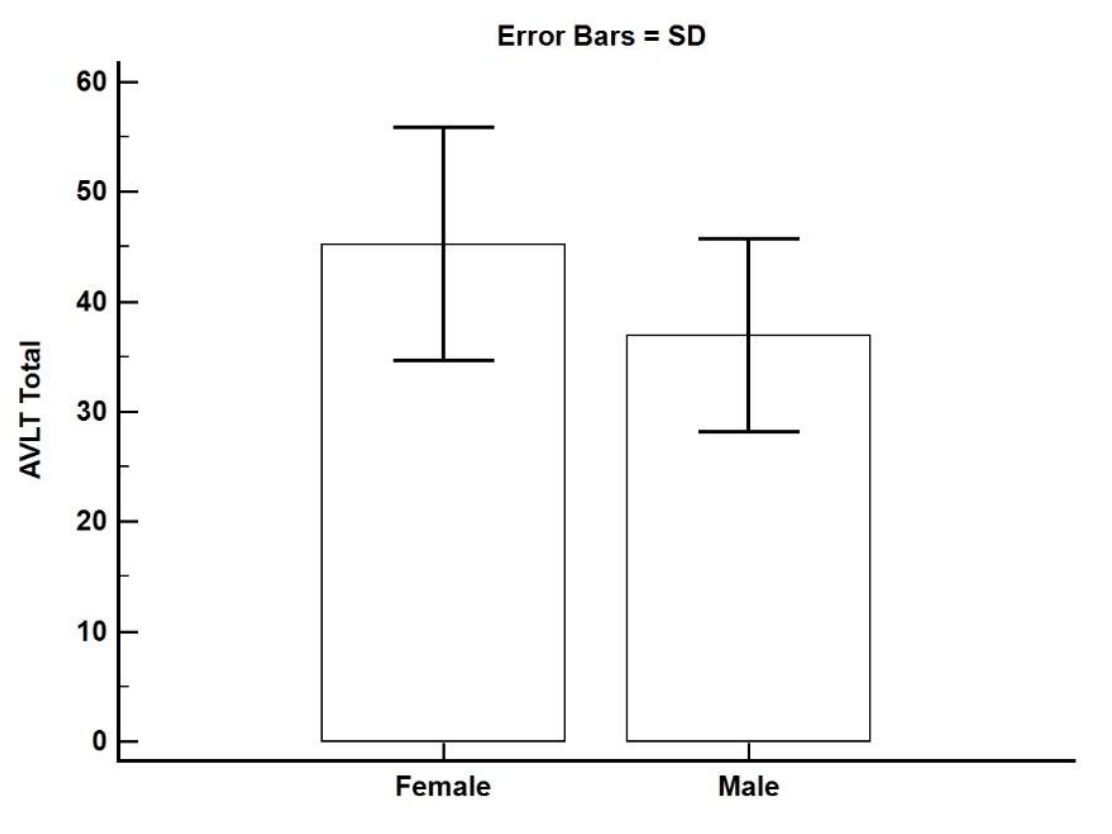
Standard Deviation vs. Standard Error

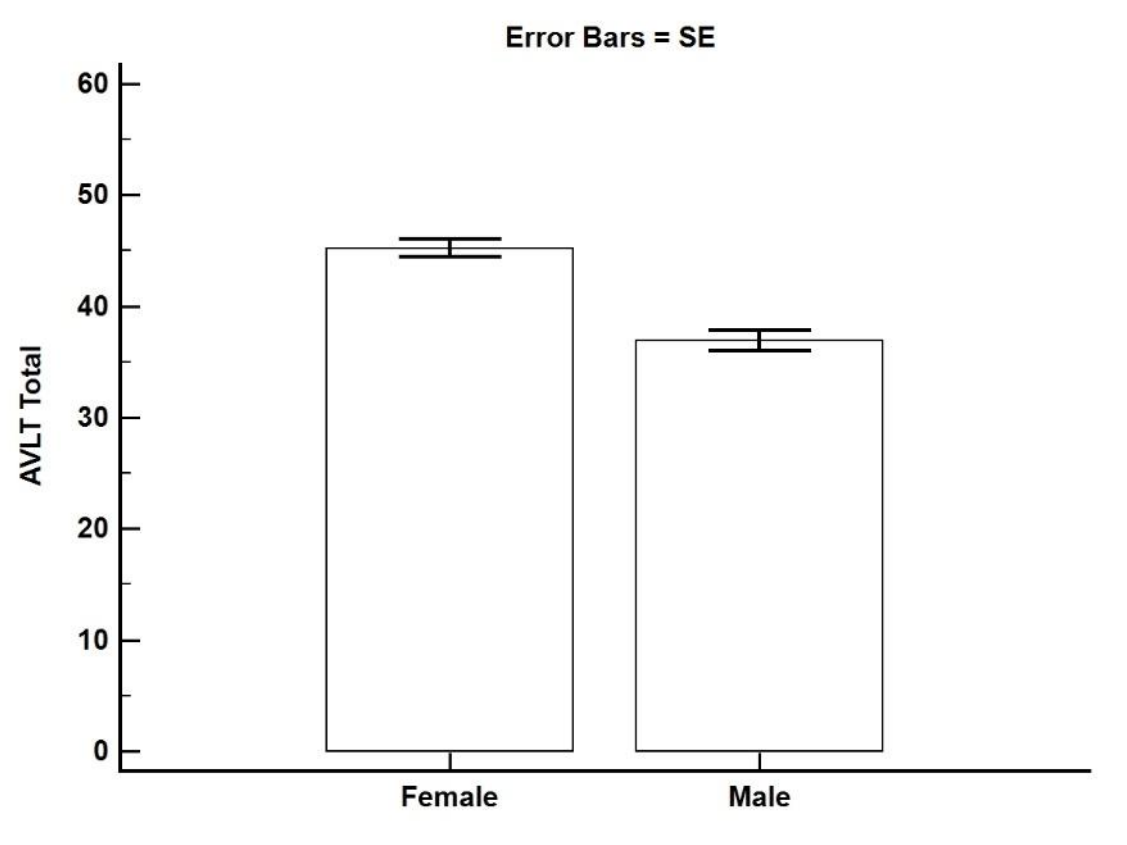
- Conceptually, standard deviation is a better representation of variability when the mean is being reported. Standard error may be acceptable for other statistics (e.g., slope coefficient).

- When describing variability relative to the mean, standard error will usually be substantially smaller than standard deviation and may not accurately reflect the true variability of data.

- This is particularly important for graphical group comparisons (e.g., bar graphs and line graphs).

Standard Deviation vs. Standard Error





Multivariate vs. Multivariable

- Multivariate: Analyses that examine multiple dependent variables simultaneously.

- Multivariable: Analyses that include multiple independent variables and are used to predict a single dependent variable.

- The two terms are not interchangeable. Multivariable analyses are more common (e.g, MMSE = age+education+sex).

Mediation vs. Moderation

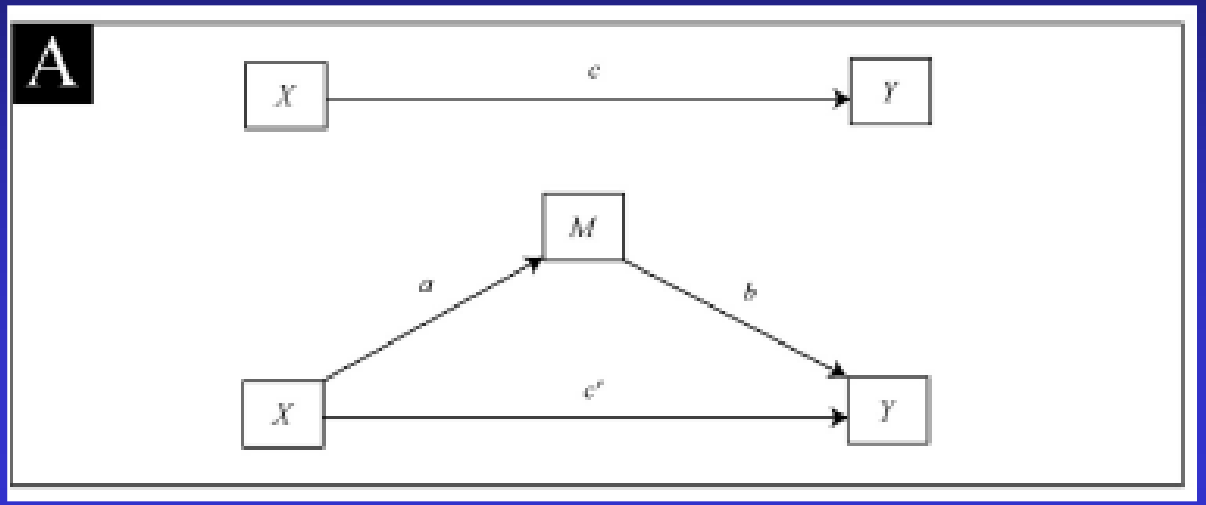
- Mediation and moderation are two concepts that describe how extraneous variables impact the relationship between the dependent and predictor variable.

- Mediation implies that a variable M lies in the causal pathway between X and Y.

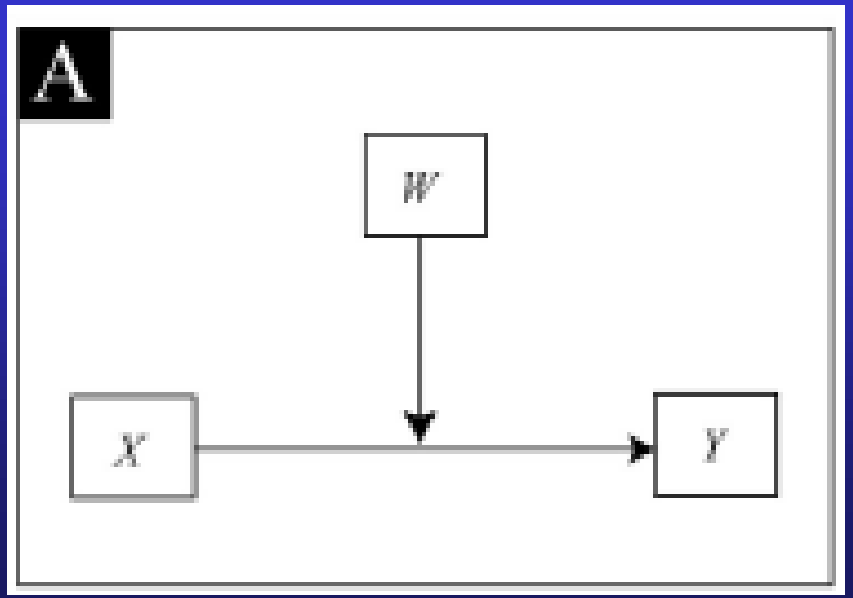
- Moderation refers to how a variable M indirectly influences the relationship between X and Y.

Mediation vs. Moderation

Mediation =



Moderation =



Hayes et al. *Behav Res Ther* 2017;98:39-57.

Moderation

- Moderating variables affect the strength and/or direction of the relation between a predictor and an outcome: enhancing, reducing, or changing the influence of the predictor.

- In regression models, the regression coefficient for the interaction term provides an estimate of the moderation effect.

Moderation in a Regression Model

- Does Florbetapir SUVR moderate the association between ischemic risk and white matter hyperintensity volume?

```
Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)     -1.1141     0.6668  -1.671  0.09581 .
hachinski        0.8310     0.8484   0.979  0.32815
suvr             1.9514     0.6092   3.203  0.00151 **
hachinski:suvr  -0.5618     0.7845  -0.716  0.47443
```

- Interaction between ischemic risk and SUVR is not significant ($p = 0.47$); SUVR does not moderate the association.

Reporting Effect Size for Mean Differences

- When reporting means, standard deviations, and p-values for two-group comparisons, please add an additional column for effect size.

- Cohen's d is the most common effect size measure for the difference in two means.

- Cohen's d Interpretation: 0.20 – 0.49=Small, 0.50 – 0.79=Medium, 0.80 or greater=Large

Cohen's d Calculation in R

- In the ClinSigMeasures package there is a function for Cohen's d.
- You just need the means and sds for the two groups.
- cohens_d (mean 1, sd1, mean2, sd2)

Cohen's d Example

	Neuritic Plaques			
Cognitive Domain	Present ($n=77$)	Absent ($n=46$)	p-value	Cohen's d
Global Cognitive Score	0.49 ± 0.29	0.75 ± 0.35	0.001	0.81
Episodic Memory	0.77 ± 0.42	1.01 ± 0.36	0.03	0.62
Semantic Memory	0.34 ± 0.46	0.65 ± 0.40	0.008	0.72
Working Memory	0.42 ± 0.50	0.61 ± 0.53	0.14	0.37
Perceptual Speed	0.51 ± 0.56	0.77 ± 0.63	0.13	0.44
Visuospatial	0.39 ± 0.52	0.73 ± 0.68	0.004	0.56

- Cohen's d for Global Cognitive Score
- cohens_d (0.75, 0.35, 0.49, 0.29)

```
> cohens_d (0.75, 0.35, 0.49, 0.29)
Cohens d = 0.808952808673172
```

Malek-Ahmadi M, Perez SE, Chen K, Mufson EJ. Neuritic and Diffuse Plaque Associations with Memory in Non-Cognitively Impaired Elderly. *J Alzheimers Dis*. 2016;53(4):1641-1652. doi:10.3233/JAD-160365

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